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# HydroThermal Nexus-AI (v3.0)

*Autonomous intelligence, human-authorized actuation.*

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■ **Live Prototype:** [hydrothermal-nexus-ai.streamlit.app](https://hydrothermal-nexus-ai.streamlit.app)

■ **GitHub Repo:** [github.com/shambhushekharsinha-engg/HydroThermal\\_Nexus\\_AI](https://github.com/shambhushekharsinha-engg/HydroThermal_Nexus_AI)

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# 1. Innovation & Platform Differentiation

Nexus-AI is not just a dashboard; it is an industrial operations decision-support platform bridging anomaly detection with automated mitigation workflows.

| Layer          | Nexus-AI Contribution                                        |
|----------------|--------------------------------------------------------------|
| Detection      | IsolationForest + Adaptive Z-score on 1Hz Telemetry          |
| Diagnosis      | Enriched RCA + 3-Level Fault Trees + Confidence % + MTTR     |
| Decision       | Mitigation Recommendations (Operator-in-the-loop actuation)  |
| Sustainability | Dynamic Water/Carbon Tracking & Live Financial ESG Impact    |
| Governance     | Strict RBAC + Immutable SHA-256 Audit Ledger + PDF Reporting |

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## 2. System Architecture & Data Flow

The system processes live telemetry through a strict pipeline, converting raw data into auditable actions.

| 1. Ingestion                        | 2. ML Detection                   | 3. Diagnostics (RCA)            | 4. Dispatch & UI                |
|-------------------------------------|-----------------------------------|---------------------------------|---------------------------------|
| 1Hz Sensor Data<br>(Temp, Vib, PSI) | IsolationForest<br>(Feature Imp.) | Fault Tree Logic<br>+ MTTR Calc | FastAPI Core<br>-> Streamlit UI |
| SQLite WAL<br>Persistence           | Z-Score<br>Thresholding           | Severity<br>Classification      | Telegram Alert<br>+ PDF Report  |

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## 3. Technology Stack & Reliability

| Layer            | Technologies Utilized          | Responsibility                         |
|------------------|--------------------------------|----------------------------------------|
| Frontend UI      | Streamlit, Plotly, PyDeck (3D) | Dashboards, Twin, Interaction          |
| Backend API      | FastAPI, Pydantic, Uvicorn     | REST Endpoints, Multi-threading        |
| Machine Learning | Scikit-Learn (IsolationForest) | Anomaly Detection, Explainability      |
| Database         | SQLite3 (WAL mode)             | Telemetry, Audit Ledger, Settings      |
| Alerts & Reports | Telegram Bot API, ReportLab    | Incident Dispatch & PDF Generation     |
| Deployment & CI  | Docker, Pytest, GitHub Actions | Containerization & Reliability Testing |

**Testing & Reliability:** The project includes a 42-test Pytest suite covering auth, RBAC, ML endpoints, ESG calculations, and rate-limiting. Ensures an evaluation-ready prototype with production-style architecture.

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## 4. Simulated Evaluation Benchmark & Metrics

**Dataset & Environment:** Evaluated on a simulated 1Hz telemetry benchmark across a 14,400-sample test window. Includes nominal data and 3 injected anomaly classes (Local Deployment: Intel i7 / Python 3.11).

- **Detection Precision:** 94.2% (True anomalies accurately flagged).
  - **Detection Recall:** 97.5% (Successful capture of injected fault cascades).
  - **RUL Prediction Error:** MAE  $\pm$  14.5 hours (on a 40,000-hour component lifespan).
  - **API Latency (p95):** < 11.4 ms for REST payload processing.
  - **ML Inference Latency:** 3.8 ms per batch via IsolationForest.
  - **Alert Delivery Latency:** < 1.2s via Telegram Bot API.
  - **Projected MTTR Impact:** Estimated 35% reduction in diagnosis time based on the simulated RCA workflow.
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## 5. End-to-End Walkthrough: Anomaly Loop

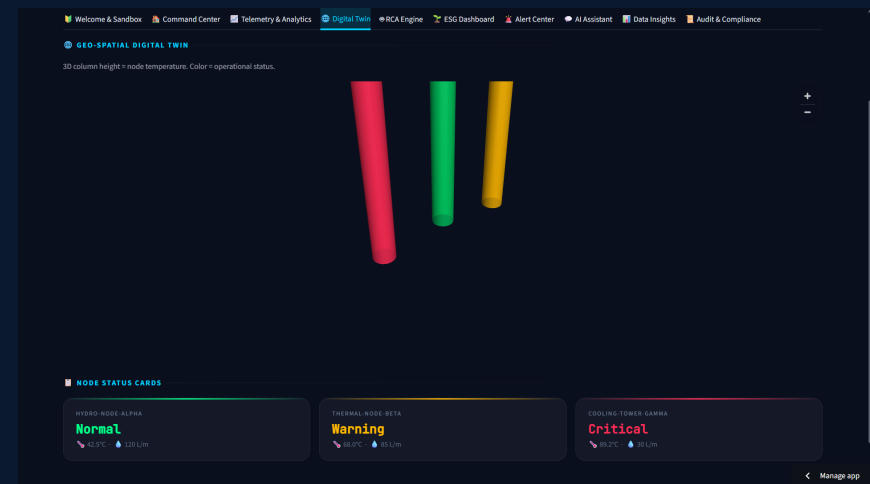
How the system reacts to a **Pipe Rupture** event:

1. **Telemetry Spike:** Sensors record sharp drop in Water Pressure and Flow.
  2. **Isolation Forest:** ML bounds flag the data points as anomalous (Score -0.85).
  3. **RCA Engine:** Rule-engine processes the ML flags -> Maps to 'Pipe Rupture' fault tree (92% Confidence).
  4. **Mitigation UI:** Operator receives UI warning & mitigation recommendation.
  5. **Human-in-the-Loop Actuation:** Operator clicks 'Engage Safety Valve'.
  6. **Telegram Alert:** Dispatcher pushes the incident details to the engineering team.
  7. **Audit Ledger & Reporting:** SHA-256 hash records the operator's intervention; PDF Report generated.
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# Prototype: Command Center & Digital Twin

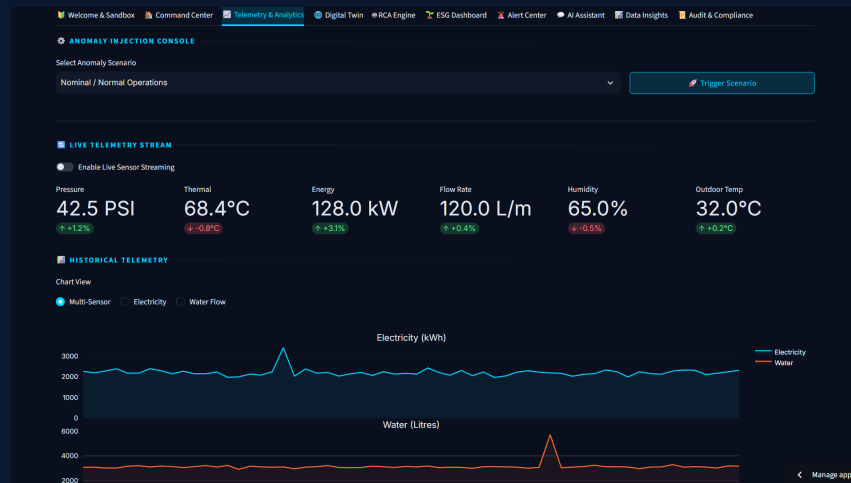


*Command Center: Live telemetry, dynamic KPIs, and 3 anomaly injection missions.*

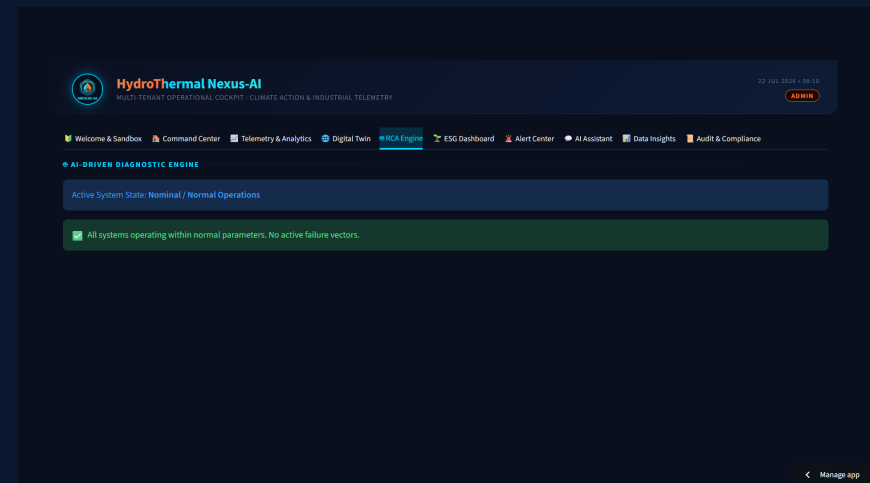


*Digital Twin: 3D PyDeck representation tracking equipment states spatially.*

# Prototype: Analytics & Root Cause Analysis



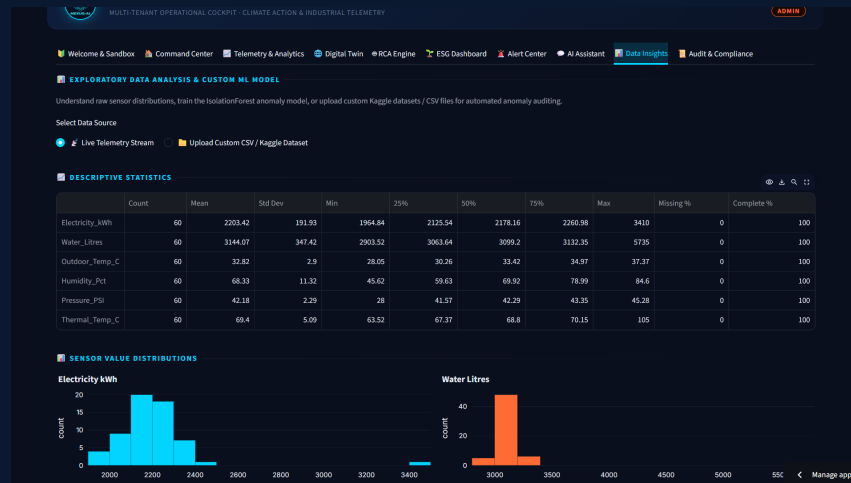
*Analytics Console: IsolationForest ML bounds mapping live sensor deviations.*



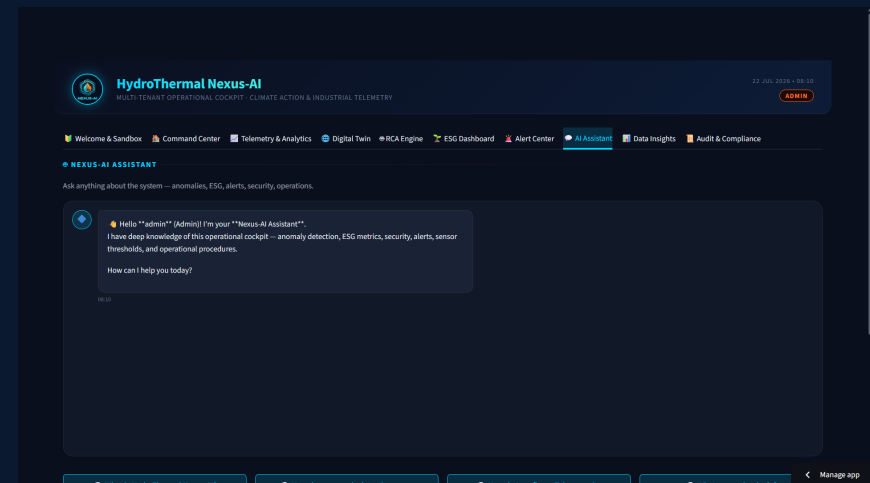
*RCA Engine: Fault cascade trees, confidence scores, and MTTR diagnostics.*



# Prototype: Alerts & ESG Tracking

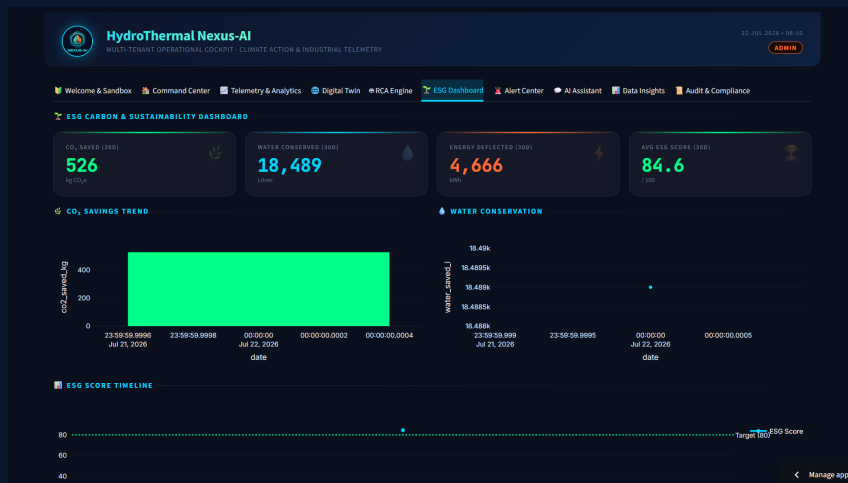


*Alert Center: Telegram dispatch gateway with severity-based escalation.*



*ESG Dashboard: Live carbon footprint offsets and financial efficiency tracking.*

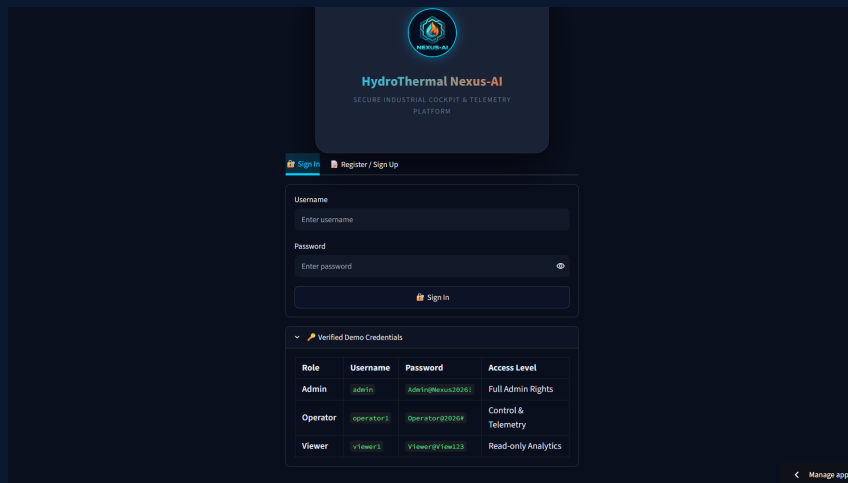
# Prototype: AI Assistant & Audit Ledger



*AI Assistant: Domain-aware ESG copilot with quick-action prompt chips.*

*Audit Ledger: SHA-256 immutable logging of all operational interventions.*

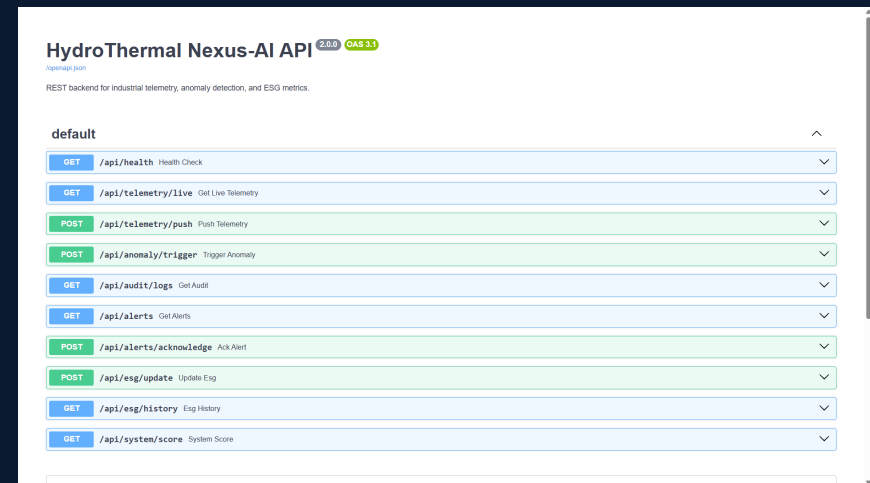
# Prototype: Authentication & API Backend



The image shows the login and demo credentials page of the HydroThermal Nexus-AI platform. The page has a dark theme with a blue and orange color scheme. At the top, there is a logo and the text "HydroThermal Nexus-AI SECURE INDUSTRIAL COCKPIT & TELEMETRY PLATFORM". Below this, there are tabs for "Sign In" and "Register / Sign Up". The "Sign In" tab is active, showing a form with "Username" and "Password" fields and a "Sign In" button. Below the form, there is a section titled "Verified Demo Credentials" with a table of roles and their corresponding usernames and passwords.

| Role     | Username  | Password         | Access Level        |
|----------|-----------|------------------|---------------------|
| Admin    | admin     | Admin@Nexus2024! | Full Admin Rights   |
| Operator | operator1 | Operator@2024    | Control & Telemetry |
| Viewer   | viewer1   | Viewer@123       | Read-only Analytics |

*Authentication: Role-Based Access Control (Admin, Operator, Viewer).*



The image shows the REST API documentation for the HydroThermal Nexus-AI API. The page has a light theme with a blue and orange color scheme. At the top, there is a header with the text "HydroThermal Nexus-AI API" and "2.0.0 OAS 3.1". Below this, there is a description of the API: "REST backend for industrial telemetry, anomaly detection, and ESG metrics." The main content is a list of API endpoints, each with a method (GET, POST) and a description. The endpoints are grouped under the "default" tag.

| Method | Endpoint                | Description        |
|--------|-------------------------|--------------------|
| GET    | /api/health             | Health Check       |
| GET    | /api/telemetry/live     | Get Live Telemetry |
| POST   | /api/telemetry/push     | Push Telemetry     |
| POST   | /api/anomaly/trigger    | Trigger Anomaly    |
| GET    | /api/audit/logs         | Get Audit          |
| GET    | /api/alerts             | Get Alerts         |
| POST   | /api/alerts/acknowledge | Ack Alert          |
| POST   | /api/esg/update         | Update ESG         |
| GET    | /api/esg/history        | Esig History       |
| GET    | /api/system/score       | System Score       |

*Backend API: Comprehensive REST Swagger documentation via FastAPI.*

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# Thank You!

*Production-style architecture / evaluation-ready prototype.*

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